

Year 7 · Chapter 12 · Ratios, rates and time: answers (teacher copy)

12.01 Ratios (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 3 : 5
2. Ex 1b: 4 : 6
3. Ex 2a: 38 : 17
4. Ex 2b: 17 : 55
5. Ex 3a: 2 : 1
6. Ex 3b: No: the order matters

Easy · Set A

1. 1 : 5
2. 5 : 3
3. 4 : 5
4. 7 : 5
5. 2 : 4
6. 3 : 7

Easy · Set B

1. 13 : 12
2. 12 : 13
3. 13 : 25
4. 12 : 25
5. 1 : 25
6. 25 : 1

Medium · Set C

1. 7 : 4
2. 4 : 11
3. 9 : 5
4. 5 : 16
5. 3 : 2
6. 4 : 9

Challenging · Set D

1. $\frac{1}{5}$
2. 8
3. 3 shaded, 2 unshaded in each 5 (6 : 4)
4. yes, e.g. 2 : 3 as $\frac{2}{3}$ of the second amount

Extension

1. $\frac{3}{10}$
2. 30%

Exit ticket

1. Level 1: 6 : 5
2. Level 2: 5 : 11
3. Level 3: 6 : 7

12.02 Equivalent and simplified ratios

(booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 8 : 12
2. Ex 1b: 6 : 9 : 12
3. Ex 2a: 4 : 7
4. Ex 2b: 3 : 1 : 4
5. Ex 3a: 2 : 1
6. Ex 3b: 1 : 3

Easy · Set A (quick drill)

1. 12
2. 10
3. 15
4. 6
5. 9, 12
6. 3
7. 2 : 3
8. 2 : 3
9. 1 : 4
10. 3 : 2
11. 2 : 3 : 5
12. 3 : 4

Easy · Set B

1. 4 : 7
2. 1 : 3
3. 1 : 6
4. 5 : 9
5. 3 : 1 : 4
6. 1 : 5 : 20

Medium · Set C

1. 2 : 1
2. 10 : 1
3. 5 : 1
4. 1 : 20
5. 1 : 5
6. 1 : 2

Challenging · Set D

1. 1 : 3
2. 4 : 7
3. 48 : 5
4. 3 : 5

Extension

1. 6 : 5
2. 1 : 3 : 6

Exit ticket

1. Level 1: 21
2. Level 2: 2 : 5
3. Level 3: 3 : 20

12.03 Ratio problems (booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: 100
2. Ex 1b: \$45 000
3. Ex 2a: \$20 and \$40
4. Ex 2b: \$200 and \$250
5. Ex 3a: 1.5 km
6. Ex 3b: 12 kg

Easy · Set A (quick drill)

1. 16
2. 12
3. 10
4. 48
5. 20
6. 45
7. \$30, \$30
8. \$20, \$40
9. \$15, \$45
10. \$24, \$36
11. \$10, \$50
12. \$35, \$25

Easy · Set B

1. \$20, \$40
2. \$40, \$60
3. \$20, \$25
4. 45, 27
5. \$200, \$800
6. 8, 16, 24 kg

Medium · Set C

1. 100
2. \$45 000
3. 15 L
4. \$200, \$250
5. 1.5 km
6. 12 kg

Challenging · Set D

1. 320
2. Sam \$36, Jo \$60
3. 750 g
4. 30 cm by 45 cm

Extension

1. 12
2. A \$28, B \$14, C \$42

Exit ticket

1. Level 1: 35
2. Level 2: \$30 and \$50
3. Level 3: 1 L

12.04 Rates (booklet p. 15)**WE DO examples (teacher)**

1. Ex 1a: 3000 voters/politician
2. Ex 1b: 200 sheep/ha
3. Ex 2a: \$0.80/ticket (or 1.25 tickets/\$)
4. Ex 2b: 270 stitches/ball
5. Ex 3a: 80 km/h
6. Ex 3b: 1500 m/min

Easy · Set A (quick drill)

1. 3000 voters/politician
2. 200 sheep/ha
3. \$0.80/ticket
4. 270 stitches/ball
5. \$15/h
6. 80 km/h
7. \$5/kg
8. 60 km/h
9. 5 L/min
10. \$12/h
11. 60 words/min
12. \$0.80/pen

Easy · Set B

1. \$5/kg
2. 15 L/min
3. 90 words/min
4. 1.2 beats/s
5. \$0.90/L
6. 14 km/L

Medium · Set C

1. 80 km/h
2. 5 L/min
3. \$22/h
4. 800 km/h
5. 25 pages/min
6. 5 m/s

Challenging · Set D

1. 1500 m/min
2. 18 km/h
3. 30 c/min
4. 20 m/s (= 72 km/h)

Extension

1. 54 km/h
2. 10.8 m/h

Exit ticket

1. Level 1: \$9/kg
2. Level 2: 75 km/h
3. Level 3: 1000 m/min

12.05 Best buys (booklet p. 19)**WE DO examples (teacher)**

1. Ex 1a: \$1.30
2. Ex 1b: \$1.11; Chow
3. Ex 2a: Dairy Farm (\$2.85/L vs \$3.12/L)
4. Ex 2b: 3 L (\$2.20/L vs \$2.30/L)
5. Ex 3a: e.g. it may go off or they cannot store it
6. Ex 3b: 5 for \$6.50 (\$1.30 vs \$1.40)

Easy · Set A (quick drill)

1. \$0.60
2. \$0.65
3. \$3.50/kg
4. \$1.70/L
5. \$0.75
6. \$1.40/m
7. \$1.50
8. \$4.50/kg
9. \$2.40/L
10. \$0.45
11. \$2.50/m
12. \$1.60

Easy · Set B

1. \$1.30
2. \$1.11
3. \$1.20
4. \$0.85
5. \$0.80
6. \$0.95

Medium · Set C

1. Chow
2. 1 L (\$2.85/L vs \$3.12/L)
3. 3 L
4. 800 g (\$0.70 vs \$0.75 per 100 g)
5. 10 rolls (\$0.75 vs \$0.80)
6. 1.25 L (\$3.60/L vs \$4/L)

Challenging · Set D

1. 5 kg (\$2.10/kg)
2. 5 c per 100 g
3. no: waste, storage, use-by dates
4. own answer

Extension

1. 1 L: \$1.35 vs \$1.50 per 100 mL, 15 c cheaper
2. \$3.00

Exit ticket

1. Level 1: \$4.20/kg
2. Level 2: 2 L (\$2.70/L)
3. Level 3: 1 kg (\$0.90/100 g)

12.06 Rate problems (booklet p. 23)

WE DO examples (teacher)

1. Ex 1a: \$853.10
2. Ex 1b: 36.3 L
3. Ex 2a: 7.5 ha
4. Ex 2b: 200 km
5. Ex 3a: 3 h
6. Ex 3b: 17.28 L

Easy · Set A (quick drill)

1. \$853.10
2. 200 km
3. \$14
4. 120 L
5. 720 words
6. \$71.60
7. \$60
8. 180 km
9. \$14
10. 60 L
11. 4 kg
12. 4 h

Easy · Set B

1. 36.3 L
2. 7.5 ha
3. 3 h
4. 15 min
5. 12 h
6. 25 min

Medium · Set C

1. 17.28 L
2. \$417.50
3. 18 km
4. 3.36 kg
5. 103 680
6. 27 L

Challenging · Set D

1. 30 min
2. \$66.60
3. 10 min
4. 100 m in 12.5 s (8 m/s vs 6.67 m/s)

Extension

1. 360 km
2. after 2 hours

Exit ticket

1. Level 1: \$147
2. Level 2: 3 h
3. Level 3: 31.25 L

12.07 Travel graphs (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 22 km
2. Ex 1b: 2
3. Ex 2a: 3 p.m.
4. Ex 2b: 1 p.m. to 2 p.m.
5. Ex 3a: 4 km/h
6. Ex 3b: faster

Easy · Set A

1. 22 km
2. 2
3. 3 p.m.
4. 1 p.m. to 2 p.m. (5 km/h)

Easy · Set A2 (quick drill)

1. 30 km/h
2. 4 km/h
3. 100 km/h
4. 10 km/h
5. 0 km/h (stopped)
6. 60 km/h

Easy · Set B

1. 300 km
2. 1 hour
3. 80 km/h
4. 40 km/h

Medium · Set C

1. travel graph

Challenging · Set D

1. stopped (not moving)
2. returning to the start
3. e.g. drove 160 km, stopped for lunch for 1 h, then drove on
4. 60 km/h

Extension

1. graph
2. 10 a.m.

Exit ticket

1. Level 1: stopped
2. Level 2: 15 km/h
3. Level 3: 3 km

12.08 Time (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: 10 h
2. Ex 1b: 3 h; 12 min
3. Ex 2a: 5 h 16 min
4. Ex 2b: 3 min 28 s
5. Ex 3a: 10 h 5 min
6. Ex 3b: 29 min

Easy · Set A (quick drill)

1. 10 h
2. 3 h
3. 4 h
4. 8 h
5. 2 h
6. 12 h
7. 120 min
8. 1 h 30 min
9. 180 s
10. 2 min 30 s
11. 30 min
12. 24 h

Easy · Set B

1. 5 h 16 min
2. 3 min 28 s
3. 150 min
4. 195 min
5. 24 min
6. 1440 min

Medium · Set C

1. 10 h 5 min
2. 4 h 42 min
3. 2 h 9 min
4. 29 min
5. 7 h
6. 1 h 30 min

Challenging · Set D

1. 9:53 p.m.
2. 10 min 55 s
3. 15 s
4. 2 h 51 min

Extension

1. age $\times$ 8760 (approx.)
2. 11 days 13 h 46 min 40 s

Exit ticket

1. Level 1: 2 h 30 min
2. Level 2: 4 h 15 min
3. Level 3: 2 h 25 min

12.09 12-hour and 24-hour time (booklet p. 37)**WE DO examples (teacher)**

1. Ex 1a: 3:50 p.m.
2. Ex 1b: 1:48 a.m.
3. Ex 2a: 12:30 p.m.
4. Ex 2b: 14:10
5. Ex 3a: 00:19
6. Ex 3b: no confusion between a.m. and p.m.

Easy · Set A (quick drill)

1. 1:48 a.m.
2. 6:24 a.m.
3. 3:50 p.m.
4. 12:30 p.m.
5. 11:05 p.m.
6. 12:40 a.m.
7. 2:30 p.m.
8. 9:15 a.m.
9. 12:20 a.m.
10. 19:45
11. 11:05
12. 00:10

Easy · Set B

1. 14:10
2. 05:33
3. 00:19
4. 13:50
5. 23:45
6. 12:05

Medium · Set C

1. 3:45 p.m.
2. 35 min
3. the 15:10 bus (arrives 15:31)
4. every 30 minutes

Challenging · Set D

1. 02:15, 2:15 a.m.
2. 7 h 45 min
3. 23:40
4. 7:15 p.m. (19:15)

Extension

1. 05:35
2. 515 min

Exit ticket

1. Level 1: 5:25 p.m.
2. Level 2: 20:05
3. Level 3: 00:45

12.10 Time differences (booklet p. 41)**WE DO examples (teacher)**

1. Ex 1a: 4 h 20 min
2. Ex 1b: 7 h 35 min
3. Ex 2a: 6 h 30 min
4. Ex 2b: 30 min
5. Ex 3a: 19 years 2 months 13 days
6. Ex 3b: 18 days

Easy · Set A (quick drill)

1. 4 h 20 min
2. 6 h 55 min
3. 6 h 30 min
4. 2 h 25 min
5. 5 h 35 min
6. 2 h 25 min
7. 3 h 30 min
8. 1 h 15 min
9. 3 h 45 min
10. 2 h 40 min
11. 3 h
12. 35 min

Easy · Set B

1. 7 h 35 min
2. 30 min
3. 11 h 55 min
4. 7 h 45 min
5. 7 h 40 min
6. 5 h 45 min

Medium · Set C

1. 18 days
2. 45 days
3. 8 weeks 4 days
4. 19 years 2 months 13 days
5. own answer
6. 1 day 6 h 35 min

Challenging · Set D

1. 4 h 55 min
2. 33 days
3. 8 years 10 months
4. \$12.80 (4 part hours)

Extension

1. 10:00 a.m.
2. 07:00 the next day (Sydney)

Exit ticket

1. Level 1: 2 h 45 min
2. Level 2: 2 h 40 min
3. Level 3: 15 days

12.11 Timetables (booklet p. 45)

WE DO examples (teacher)

1. Ex 1a: 8:33 a.m.
2. Ex 1b: 2 h 16 min
3. Ex 2a: 7:40 a.m.
4. Ex 2b: 1 h 13 min
5. Ex 3a: the 7:40 a.m. train (arrives 9:56 a.m.)
6. Ex 3b: no confusion between a.m. and p.m.

Easy · Set A

1. 8:21 a.m.
2. 2 h 16 min
3. 9:25 a.m.
4. 1 h 23 min

Easy · Set A2 (quick drill)

1. 45 min
2. 1 h 25 min
3. 1 h 35 min
4. 2 h 16 min
5. 35 min
6. 2 h 9 min

Easy · Set B

1. 8:33 a.m.
2. 2 h 16 min
3. 7:40 a.m.
4. 1 h 13 min
5. 22:01
6. 1 h 35 min

Medium · Set C

1. 27 min
2. 16:18, reaching the Beach at 16:45
3. the 15:40 bus (Shops 16:01); the 16:10 bus reaches the Shops at 16:31, too late
4. 4:10, 4:18, 4:31, 4:45 p.m.

Challenging · Set D

1. 06:50 → 07:15, 07:30 → 07:55, 08:10 → 08:35, 08:50 → 09:15, 09:30 → 09:55
2. own timetable

Extension

1. 8:33 a.m.
2. 10:08 a.m.

Exit ticket

1. Level 1: 10:30 a.m.
2. Level 2: 21 min
3. Level 3: the 6:05 a.m. train